NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-19 | Date: 17.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	D	A	B	B	C	A	B	D	B
11	12	13	14	15	16	17	18	19	20
C	B	D	C	C	C	C	A	D	B
21	22	23	24	25					
3	16	31	400	673					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	D	B	A	A	C	B	B	D	D
36	37	38	39	40	41	42	43	44	45
D	A	A	B	B	D	B	A	A	A
46	47	48	49	50					
108	2	4	16	1200					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
D	D	B	A	D	C	D	A	C	B
61	62	63	64	65	66	67	68	69	70
A	B	A	D	D	D	B	B	B	D
71	72	73	74	75					
12	48	2890	119	5					

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PHYSICS

1.	$\tan\theta = \frac{v^2}{rg} = \frac{h}{w}$ $\therefore \quad h = \frac{v^2}{rg}w$ $= \frac{(12)^2 \times 1.5}{400 \times 10}$ $= 0.054$ m $= 5.4$ cm
2.	$V = \frac{KQ}{r}$ at $P : r_1 = \sqrt{6}$ at $Q : r_2 = \sqrt{6}$ $\therefore \quad V_P = V_Q$ $\Rightarrow \quad \Delta V = 0$
3.	For minimum deviation $\mu = \frac{\sin\frac{(\delta + A)}{2}}{\sin\left(\frac{A}{2}\right)} = \cot\left(\frac{A}{2}\right)$ $\delta = \pi - 2A$
4.	$E_{avg} = -\frac{\Delta\phi}{\Delta t}$ $\Delta\phi = -BA\cos 60°$ $= -4 \times 2.5 \times 2\frac{1}{2}$ $= -10$ $E_{avg} = \frac{+10}{10} = 1$ V
5.	Viscosity of liquids is greater than gas. Surface tension decreases due to the presence of impurities.

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6.	$\Delta U = mC\Delta T$ $= 0.08 \times 0.17 \times 4.18 \times 10^3 \times 5$ $= 284$ J
7.	$[h] = [ML^2T^{-1}]$ $[L] = [ML^2T^{-1}]$ $[P] = [MLT^{-1}]$ $[FL] = [MLT^{-2}] \times [L]$ $= [ML^2T^{-2}]$
8.	Spherometer cannot measure specific rotation of liquids.
9.	$\because \ g = \frac{GM}{R^2}$ If $R' = \frac{R}{2}$ $\Rightarrow \ g' = 4g$
10.	By conservation of momentum $1000 \times 6 = 1200 \times V$ $\Rightarrow \ v = 5$ m/s
11.	As the diameter is same, number of photons will be same. $\therefore \ I' = I$
12.	$I = \frac{1}{2} \epsilon_0 E_0^2 C$ $= \frac{1}{2} \times 8.85 \times 10^{-12} \times (200)^2 \times (3 \times 10^8)$ $= 53.1$ W m^{-2}
13.	Resistance of each part $= \frac{R}{5}$ $\therefore \ R_{eq} = \left(\frac{1}{5}\right)\left(\frac{R}{5}\right) = \frac{R}{25}$
14.	For constant velocity, net force on charge particle should be zero. That is possible for A, B and D.

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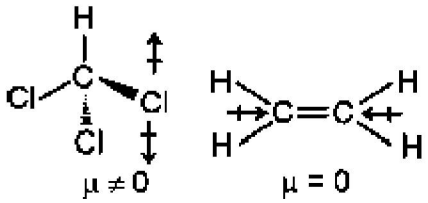
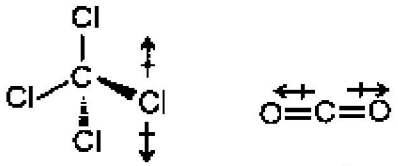
15.	$\frac{4.5}{60} = \frac{S}{40}$ $\therefore \quad S = 3\ \Omega = \frac{\rho l}{A}$ $\rho = \frac{SA}{l} = 66 \times 10^{-7}$ $\therefore \quad R = 66$
16.	$E = \frac{3}{2} K_B T$ $0.414 \times 1.6 \times 10^{-19} = \frac{3}{2} \times 1.38 \times 10^{-23} T$ $T = 3200\ \text{K}$
17.	$K = \frac{P^2}{2m}$ $\frac{P_1^2}{2m_1} = \frac{P_2^2}{2m_2}$ $\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \frac{2}{5}$
18.	$\vec{v} = \frac{d\vec{s}}{dt}$ $\vec{v} = 4t\,\hat{j}$ At $t = 1$ s $\vec{v} = 4\,\hat{j}$
19.	For 3rd orbit $R = r_0 \frac{(3)^2}{Z}$ $R' = r_0 \frac{(4)^2}{Z}$ $\therefore \quad R' = \left(\frac{16}{9}\right) R$
20.	For reverse bias $V_P < V_N$, that is true for option (4)

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21.	Force on one half of the ring = $\frac{2K\lambda Q}{r}$ $\therefore T = \frac{K\lambda Q}{r}$ $= 9\times10^{9}\times\frac{2\pi}{2\pi\times0.3}\times\frac{30\times10^{-12}}{0.3}$ $= 3$ N
22.	$I = 2(Ma^2) + M(\sqrt{2}a)^2$ $I = 4Ma^2$ $= 4 \times 1 \times 4 = 16$
23.	Apparent depth $= \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2}$ $= \frac{6}{\left(\frac{3}{2}\right)} + \frac{6}{\left(\frac{8}{5}\right)} = \frac{31}{4}$ cm
24.	$V_A = 10 - \frac{10}{3} = \frac{20}{3}$ V $V_B = 10 - 6 = 4$ V $V_A - V_B = \frac{8}{3}$ V $Q = CV_{AB} = 150\times10^{-6}\times\frac{8}{3} = 400\ \mu C$
25.	$x = 5t + \frac{1}{2}\times 3t^2 = 84$ $\therefore t = 6$ s $\vec{v} = 5\hat{i} + 6(3\hat{i} + 2\hat{j})$ $\vec{v} = 23\hat{i} + 12\hat{j}$ $v = \sqrt{673}$

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CHEMISTRY

26.	p-Nitrophenol is more acidic than m-nitrophenol and o-nitrophenol. The electron withdrawing group $-NO_2$ stabilizes phenoxide ion through –R effect at para and ortho position. $-NO_2$ group in meta position will show –I effect. Presence of $-NO_2$ group at ortho position it form hydrogen bonding with –OH group. Hence, acidity is less than p-nitrophenol order of acidic strength. p-nitrophenol > o-nitrophenol > m-nitrophenol Ethanol will not give immediate turbidity with Lucas reagent. 3° alcohol will give white turbidity immediately statement I is true but statement II is false.
27.	The molecule which has permanent dipole moment are said to be polar molecule.  $\mu \neq 0$ $\mu = 0$  $\mu = 0$ $\mu = 0$ $CHCl_3$ is polar molecule
28.	[Ar] $3d^6$ have 4 unpaired electrons which has highest magnetic moment.

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29.

P = $[FeF_6]^{3-}$

Q = $[V(H_2O)_6]^{2+}$

R = $[Fe(H_2O)_6]^{2+}$

$F^{\ominus}$ with Fe^{+3}, and H_2O with Fe^{+2} behaves as WFL and pairing does not takes place

$[FeF_6]^{3-} = 3d^5 = t_{2g}^3 e_g^2$

Number of unpaired electron = 5

$\mu = \sqrt{n(n+2)}$ BM

$= \sqrt{5(5+2)}$ BM

$= \sqrt{35}$ BM

$[V(H_2O)_6]^{2+} = 3d^3$ $\quad t_{2g}^3 eg^0$

n = 3

$\mu = \sqrt{15}$ BM

$R = \left[Fe(H_2O)_6\right]^{2+} = 3d^6 = t_{2g}^4 e_g^2$

n = 4

$\mu = \sqrt{24}$ BM

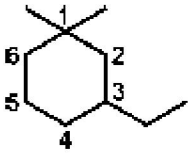
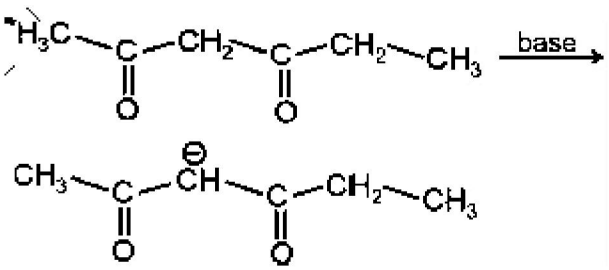
$\boxed{Q < R < P}$

30.

In S_N1 reaction carbocation is formed as intermediate. Hence both side attack is possible, racemic product is formed.

In S_N2 reaction, nucleophile attack from back side takes place and hence inversion in configuration takes place.

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31.	 Numbering is done in such a way that all locants should get lowest possible number. Name of substituent written according to alphabetical order 3-Ethyl-1,1-dimethylcyclohexane
32.	After removal of H^+ ion, negative charge is formed, more stable the negative charge more will be the acidic strength.  Due to cross conjugation of negative charge it is most stabilised anion. So compound is most acidic.
33.	Aq solution of ammonium carbonate is basic. The salt is made from weak acid (H_2CO_3) and weak base (NH_4OH), acidic or basic nature depends on value of K_a of weak acid and K_b of weak base If solution is basic K_b (weak base) > K_a (weak acid) $K_a(H_2CO_3) = 4.3 \times 10^{-7}$ $K_b (NH_4OH) = 1.8 \times 10^{-5}$ $K_b > K_a$ so solution is basic Both statements are correct
34.	 The given compound belongs to alicyclic compounds.

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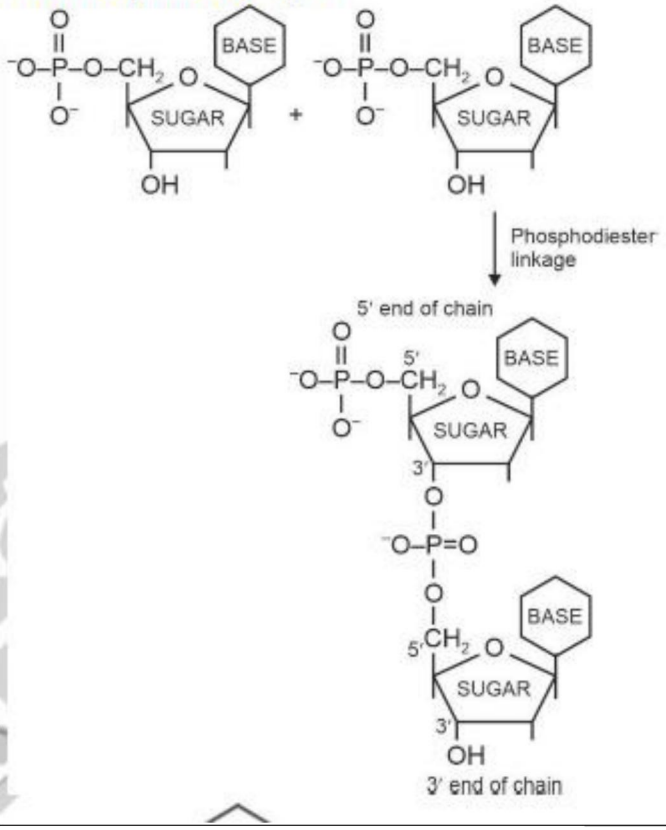
35.

$$4NaCl + K_2Cr_2O_7 + 6H_2SO_4 \longrightarrow$$

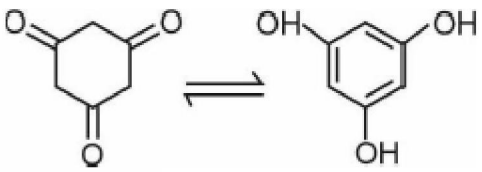
$$\underset{(B)}{2CrO_2Cl_2} + 4NaHSO_4 + 2KHSO_4 + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\substack{(C)\\ \text{Yellow}}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

36. Two nucleotides are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of pentose sugar.



37. Highest enol content will shown by

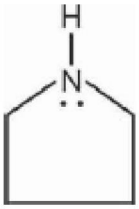
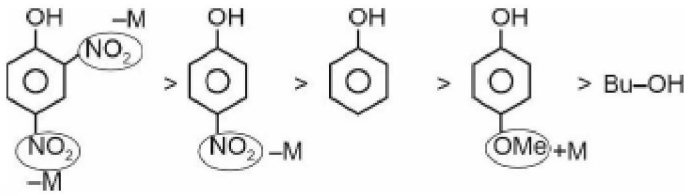


Due to formation of aromatic compound.

38. $$\underset{\text{yellow}}{PbCrO_4} \xrightarrow[NaOH]{\text{hot}} Na_2CrO_4 + \underset{\text{complex}}{[Pb(OH)_4]^{2-}}$$

The complex formed is $[Pb(OH)_4]^{2-}$ which has coordination number = 4 and complex is dianionic due to presence of (–2) charge.

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39.	The electronic configuration of Neodymium is $_{60}Nd$: $[Xe]4f^4\ 6s^2$
40.	Bronsted base are those which can donate its lone pair to H^+ ion. more easily the compound can donate its lone pair to H^+ strong will be basic character  is strongest Bronsted base due to availability of lone pair. In other option lone pair is delocalised
41.	Fluorine does not show variable oxidation state due to absence of vacant *d*-orbital.
42.	A solution of two miscible liquids showing negative deviation due to decrease in vapour pressure and increase in boiling point.
43.	*s*-block elements are highly reactive and hence can't be found in pure form in nature. ⇒ Statement II is incorrect.
44.	 Acidic strength order Bu–OH is aliphatic alcohol.
45.	Melting point of Boron is unusually high due to very strong crystalline lattice. Both (A) and (R) are correct and (R) is correct explanation of (A).
46.	$2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$ moles of $O_2 = \frac{5.6}{22.4} = 0.25$ moles of electrons = $(0.25 \times 4) = 1$ $Ag^+ + e^- \longrightarrow Ag$ moles of Ag = 1 mass of Ag = 108 gm

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47.	Comparing rate values in experiment 2 and 3 $\frac{r_3}{r_2} = 4 = \left(\frac{0.02}{0.01}\right)^x$ $\boxed{x = 2}$ Order of reaction is 2
48.	$CH_3-CH=\underset{\underset{CH_3}{|}}{C}-CH_2-CH_2-CH_3 \xrightarrow[\text{Peroxide}]{HBr}$ $CH_3-\overset{*}{\underset{\underset{Br}{|}}{C}}H-\overset{*}{\underset{\underset{CH_3}{|}}{C}}H-CH_2-CH_2-CH_3$ (A) (A) has 2 chiral carbons Possible stereoisomers = $(2)^2 = 4$
49.	Total (n^2) electrons present with $s = +\frac{1}{2}$ $\Rightarrow$ Total 16 electrons with n = 4 and $s = +\frac{1}{2}$
50.	$W = -P_{ext}(\Delta V)$ $= -80\times10^3 \frac{N}{m^2}(45-30)\times10^{-3}\,m^3$ = –80 x (15) N.m = –1200 J $q + W = \Delta E = 0$ q = +1200 J

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MATHS

51.

$${}^{n-1}C_r = \left(k^2 - 8\right)\frac{n}{r+1}\,{}^{n-1}C_r$$

$$k^2 - 8 = \frac{r+1}{n}$$

$$0 < k^2 - 8 \le 1$$

$$8 < k^2 \le 9$$

$$2\sqrt{2} < k \le 3 \text{ or } -3 \le k < -2\sqrt{2}$$

52.

A = sum of coefficient = $(1 - 3(1) + 10(1)^2)^n = 8^n$

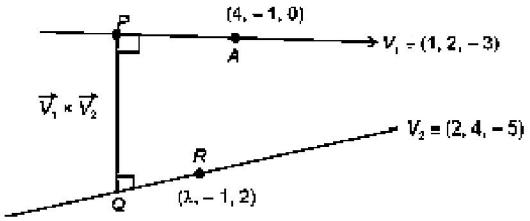
(Putting $x = 1$)

B = Sum of coefficient = $(1 + (1)^2)^n = 2^n$

$\Rightarrow 8^n = (2^3)^n = (2^n)^3 = B^3$

$\Rightarrow A = B^3$

53.



Shortest distance PQ is given by

$$\left|\frac{\vec{V}_1 \times \vec{V}_2 \cdot \overrightarrow{AB}}{|\vec{V}_1 \times \vec{V}_2|}\right| = \frac{6}{\sqrt{5}}$$

$$\vec{V}_1 \times \vec{V}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix} = 2\hat{i} - \hat{j}$$

$$\Rightarrow |\vec{V}_1 \times \vec{V}_2| = \sqrt{5}$$

$$\overrightarrow{AB} = (\lambda - 4)\hat{i} + 2\hat{k}$$

$$\Rightarrow \left|\frac{\vec{V}_1 \times \vec{V}_2}{\vec{V}_1 \times \vec{V}_2} \cdot \overrightarrow{AB}\right| = \frac{6}{\sqrt{5}} \Rightarrow \left|\frac{2(\lambda - 4)}{\sqrt{5}}\right| = \frac{6}{\sqrt{5}}$$

$\Rightarrow |\lambda - 4| = 3$

$\Rightarrow \lambda - 4 = 3$ or -3

$\lambda = 1$ or 7

$\Rightarrow$ Sum of $\lambda = 8$

54.

Let P (7, –2, 11)

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and line $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$

Let General point be 'Q'

$\therefore \quad Q \equiv (\lambda + 6, 4, 3\lambda + 8) \quad \ldots(I)$

$\therefore \quad \overrightarrow{PQ} = (1-\lambda, -6, -3\lambda - 3)$

As we have to find distance along

$\frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$

$\therefore \quad \frac{1-\lambda}{2}=\frac{-6}{-3}=\frac{-3\lambda-3}{6}$

$\frac{1-\lambda}{2}=2=\frac{-3\lambda-3}{6}$

$\therefore \quad \frac{1-\lambda}{2}=2$

$1-\lambda=4$

$\lambda=-3$

Putting in (I)

$Q \equiv (3, 4, -1)$

and $P \equiv (7, -2, 11)$

$\therefore \quad PQ = \sqrt{(3-7)^2+(4+2)^2+(-1-11)^2}$

$= \sqrt{(-4)^2+(6)^2(-12)^2}$

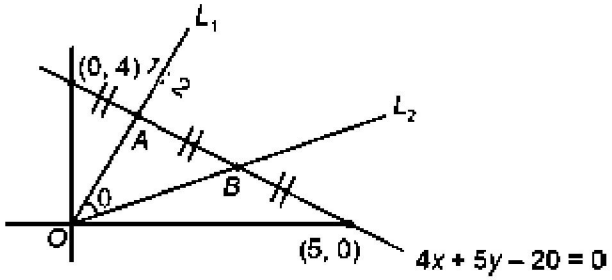
$= \sqrt{16+36+144}$

$= \sqrt{196}$

$= 14$

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55.



$$A_x \equiv \frac{5(1)+2(0)}{3} = \frac{5}{3}$$

$$\Rightarrow A \equiv \left(\frac{5}{3}, \frac{8}{3}\right)$$

$$A_y = \frac{0(1)+2(4)}{3} = \frac{8}{3}$$

Similarly, $B \equiv \left(\frac{10}{3}, \frac{4}{3}\right)$

$\Rightarrow$ Slope of $AO = \frac{8}{5}$

Slope of $OB = \frac{2}{5}$

$$\tan\theta = \left|\frac{\frac{8}{5}-\frac{2}{5}}{1+\frac{16}{25}}\right| = \left|\frac{\frac{6}{5}}{41/25}\right| = \frac{30}{41}$$

56.

$$\sum_{\forall k<j} a_k \cdot a_j = \left(\frac{\sum_{k=1}^{10}\sum_{j=1}^{10} a_k a_j - \sum_{k=1}^{10} a_k a_k}{2}\right)$$

$$\sum_{k=1}^{10}\sum_{j=1} a_k a_j = (a_1 + a_2 + \ldots + a_{10})(a_1 + a_2 + \ldots + a_{10})$$

$$= (50)(50) = 2500$$

$$\Rightarrow \sum_{k=1}^{10} a_k^2 = 2500 - 2\sum_{\forall k<j} a_k a_j$$

$$= 2500 - 2 \times 1100 = 300$$

Standard deviation (σ)

$$\Rightarrow \sigma = \sqrt{\frac{\sum a_x^2}{10} - \left(\frac{\sum a_k}{10}\right)^2}$$

$$= \sqrt{\frac{300}{10} - \left(\frac{50}{10}\right)^2} = \sqrt{30-25} = \sqrt{5}$$

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57.	Largest prime factors of two different natural numbers can be same so it is many one. For example $f(6) = 3$, $f(12) = 3$ $f(10) = 5$, $f(15) = 5$ As range of $f(x)$ is set of prime number so range of $f(x)$ is proper subset of co-domain so f is into.
58.	$S = \{1, 2, 3, \ldots, 10\}$ and $R = \{(A, B) : A \cap B \neq \phi;$ $A, B \in M\}$ $A \in M$ $A \cap A = A \neq \phi$ So, R is not reflexive If $(A, B) \in R$ $\Rightarrow \quad A \cap B = \phi$ $\Rightarrow \quad (B, A) \in R$ So, R is symmetric If $(A, B), (B, C) \in R \Rightarrow A \cap B = \phi$ and $B \cap C = \phi$ $\Rightarrow \quad A \cap C$ is not necessary ϕ. Hence, $A \cap C$ not necessarily belongs to R. So, R is not transitive relation. eg:- $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{2, 5\}$ here, $A \cap B = \phi$, $B \cap C = \phi$ but $A \cap C = \{2\} \neq \phi$ So, option (3) is correct.
59.	$\frac{dx}{dt} + ax = 0$

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$\Rightarrow \int \frac{1}{x} dx = -\int a dt$

$\Rightarrow \ln|x| = -at + c$

$\Rightarrow x = \pm e^{c} \cdot e^{-at} = \lambda e^{-at}$

$\Rightarrow x(0) = 2 \Rightarrow \lambda = 2$

$\Rightarrow x = 2e^{-at}$ ...(1)

Now, $\frac{dy}{dt} + by = 0$

$\Rightarrow \int \frac{1}{y} dy = -\int b dt \Rightarrow \ln(y) = -bt + c$

$\Rightarrow y = \pm e^{c} \cdot e^{-bt} = \mu \cdot e^{-bt}$

$y(0) = 1$

$\therefore \mu = 1$

$\Rightarrow y = e^{-bt}$

$3y(1) = 2x(1)$

$3e^{-b} = 2(2e^{-a}) = 4e^{-a}$

$e^{-b+a} = \frac{4}{3}$

$\Rightarrow x(t) = y(t)$

$\Rightarrow e^{-bt} = 2e^{-at}$

$\Rightarrow e^{-bt+at} = 2$

$\Rightarrow e^{(-b+a)t} = 2$

$\Rightarrow \left(\frac{4}{3}\right)^{t} = 2$

$\Rightarrow t \ln\frac{4}{3} = \ln 2$

$\Rightarrow t = \ln_{\frac{4}{3}} 2$

60. $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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$$f(x)\cdot f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$= f(x + y)$

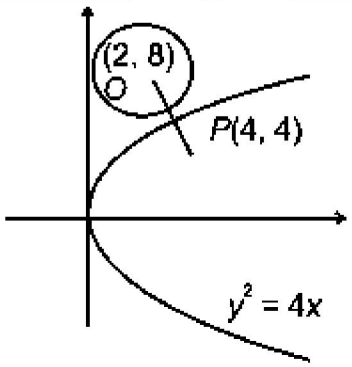
So, statement II is true.

$f(x)\, f(-x) = I$

So, $f(-x)$ is inverse of $f(x)$.

So, statement I is true.

61.



$y = mx - 2m - m^3$.

If it passes through (2, 8)

$8 = 2m - 2m - m^3$.

$\Rightarrow \quad m = -2$

Feet of normal on parabola.

$P(am^2, -2am) \equiv (4, 4)$

$d^2 = OP^2 = (4-2)^2 + (4-8)^2$

$= 4 + 16 = 20$

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62.

Given A.P.

4, 9, 14, 19, ... Upto 25 terms

Here $d_1 = 5$

$a_n = a + (n-1)d$

$a_{25} = 4 + (25-1)5$

$a_{25} = 4 + 24(5)$

$a_{25} = 124$

So, A.P. is

4, 9, 14, 19, ... 124

Second A.P. is given as

3, 6, 9, 12, ... upto 37 term.

Here $d_2 = 3$

$a_n = a + (n-1)d$

$a_n = 3 + (n-1)\cdot 3$

$a_{37} = 3 + 36(3)$

$a_{37} = 111$

So, A.P. is

3, 6, 9, 12, ..., 111

L.C.M. of d_1 and $d_2 = 15$

So, A.P. of common terms becomes

9, 24, 39, ... 99

$a_n = a + (n-1)d$

$99 = 9 + (n-1)15$

$\frac{90}{15} = n - 1$

$n = 7$

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63. If $f(x)$ is continuous at $x = 3$

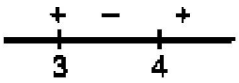
$\Rightarrow f(3) = f(3^+) = f(3^-)$

$$\Rightarrow b = \lim_{x\to 3^+} 2^{\frac{\sin(x-3)}{x-[x]}} = \lim_{x\to 3^-} \frac{a}{b}\frac{(7x-12-x^2)}{|x^2-7x+12|}$$

$$\Rightarrow \lim_{x\to 3^+} 2^{\frac{\sin(x-3)}{x-[x]}}$$

$$\Rightarrow \lim_{h\to 0} 2^{\frac{\sin h}{h}} = 2$$

$(7x - 12 - x^2) = -(x^2 - 7x + 12) = -(x-3)(x-4)$



$\Rightarrow (x^2 - 7x + 12)$ will open positive

$$\Rightarrow \lim_{x\to 3^-} \frac{a}{b}\frac{(7x-12-x^2)}{(x^2-7x+12)} = \lim_{x\to 3^-} \frac{-a}{b} = 2$$

$\Rightarrow -a = 2b \Rightarrow a = -2b = -4$

$\Rightarrow (a, b) = \{(-4, 2)\}$

$\Rightarrow$ Only 1 ordered pair

64. $$\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$$

Taking L.H.S.

Multiply and divide by $\sqrt{3+x} - \sqrt{1+x}$

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$$= \int_0^1 \frac{\sqrt{3+x} - \sqrt{1+x}}{2} dx$$

$$= \frac{1}{2}\int_0^1 (\sqrt{3+x} - \sqrt{1+x})dx$$

$$= \frac{1}{2}\left[\frac{2}{3}(3+x)^{3/2} - \frac{2(1+x)^{3/2}}{3}\right]_0^1$$

$$= \frac{1}{3}[4^{3/2} - 2^{3/2} - 3^{3/2} + 1]$$

$$= \frac{1}{3}[8 - 2\sqrt{2} - 3\sqrt{3} + 1]$$

$$= \frac{1}{3}(9 - 2\sqrt{2} - 3\sqrt{3})$$

$$= 3 - \frac{2\sqrt{2}}{3} - \sqrt{3}$$

By comparing, we get :

$a = 3$

$b = -2/3$

$c = -1$

$2a + 3b - 4c$

$= 6 + 3(-2/3) - 4(-1)$

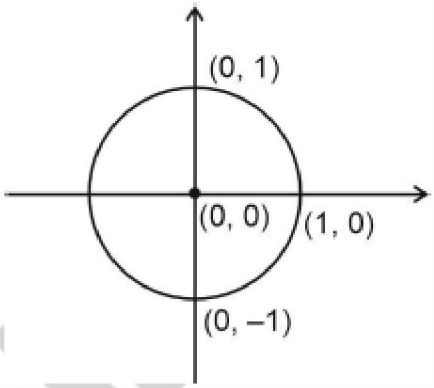
$= 6 - 2 + 4$

$= 8$

65. z will be circumcentre of the triangle with vertices i, $-i$ and 1(which is unique)

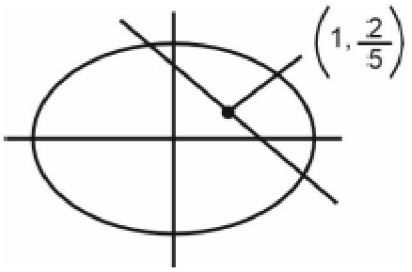
$\Rightarrow \quad z = 0 + 0i$

$\therefore$ Only one element exist in S



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66.



$$T = S_1 \Rightarrow \frac{x}{25} + \frac{y\left(\frac{2}{5}\right)}{16} = \frac{1}{25} + \frac{4}{16 \times 25}$$

$$\Rightarrow \quad \frac{x}{25} + \frac{y}{40} = \frac{5}{100}$$

$$\Rightarrow \quad 4x + \frac{5}{2}y = 5 \Rightarrow 8x + 5y - 10 = 0$$

$$\frac{x^2}{25} + \frac{y^2}{16} = 1 \Rightarrow (4x)^2 + (5y)^2 = 400$$

$$\Rightarrow \quad (4x)^2 + (8x - 10)^2 = 400$$

$$\Rightarrow \quad 4x^2 + (4x - 5)^2 = 100$$

$$\Rightarrow \quad 20x^2 - 40x - 75 = 0$$

$$\Rightarrow \quad 4x^2 - 8x - 15 = 0$$

$$l = \sqrt{(x_2 - x_1)^2 - (y_2 - y_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 - \left\{\frac{(10 - 8x_2)}{5} - \frac{(10 - 8x_1)}{5}\right\}^2}$$

$$= \sqrt{(x_2 - x_1)^2 + \frac{64}{25}(x_2 - x_1)^2}$$

$$= \sqrt{\frac{89}{25}}\sqrt{(x_2 - x_1)^2}$$

$$= \sqrt{\frac{89}{25}}\sqrt{(x_1 - x_2)^2 - 4x_1x_2}$$

$$= \frac{\sqrt{89}}{5}\sqrt{4 + 4 \cdot \frac{15}{4}} = \frac{1}{5}\sqrt{89} \times \sqrt{19}$$

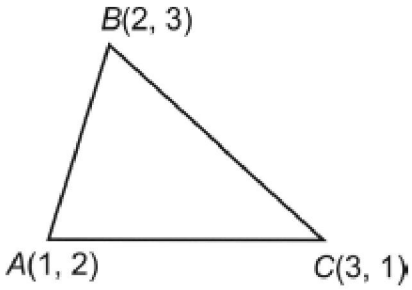
$$= \frac{\sqrt{1691}}{5}$$

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67.

Slope of $AB = 1$.

Altitude through C



$y - 1 = -1(x - 3)$

$\Rightarrow \quad x + y = 4.$

Orthocentre (a, b) lies on this

$\Rightarrow \quad a + b = 4 \qquad \ldots(i)$

Now, $I_1 = \int_a^b x \cdot \sin(4x - x^2)dx$

$$= \int_a^b (a + b - x)\sin(4(a + b - x) - (a + b - x)^2)dx$$

$$= \int_a^b (4 - x)\sin(4x - x^2)dx \qquad \{\text{using (i)}\}$$

$\Rightarrow \quad I_1 = 4 \cdot I_2 - I_1$

$\Rightarrow \quad \frac{I_1}{I_2} = 2.$

$\Rightarrow \quad 36\frac{I_1}{I_2} = 72$

68.

$$a = \lim_{x \to 0} \frac{\sqrt{1 + \sqrt{1 + x^4}} - \sqrt{2}}{x^4}$$

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and $b=\lim\limits_{x\to 0}\dfrac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$

$$a=\lim_{x\to 0}\frac{1+\sqrt{1+x^4}-2}{x^4\left(\sqrt{1+\sqrt{1+x^4}}+\sqrt{2}\right)}$$

$$=\frac{1}{2\sqrt{2}}\lim_{x\to 0}\frac{\sqrt{1+x^4}-1}{x^4}$$

$$=\frac{1}{2\sqrt{2}}\lim_{x\to 0}\frac{(1+x^4)-1}{x^4\left(\sqrt{1+x^4}+1\right)}$$

$$=\frac{1}{2\sqrt{2}}\left(\frac{1}{2}\right)=\frac{1}{4\sqrt{2}}$$

$$\Rightarrow \quad a=\frac{1}{4\sqrt{2}}$$

$$b=\lim_{x\to 0}\frac{(1-\cos^2 x)(\sqrt{2}+\sqrt{1+\cos x})}{2-(1+\cos x)}$$

$$b=\lim_{x\to 0}(1+\cos x)\left(\sqrt{2}+\sqrt{1+\cos x}\right)$$

$$\Rightarrow \quad b=2\left(2\sqrt{2}\right)$$

$$\Rightarrow \quad \boxed{b=4\sqrt{2}}$$

So, $ab^3=\dfrac{1}{4\sqrt{2}}\left(4\sqrt{2}\right)^3$

$$\boxed{ab^3=32}$$

So, option (2) is correct.

69. $\vec{a}=\hat{i}+2\hat{j}+\hat{k}, \vec{b}=3(\hat{i}-\hat{j}+\hat{k})$

$\vec{a}\cdot\vec{c}=3$ ...(1)

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$\vec{a}\times\vec{c}=\vec{b}$ ...(2)

Taking cross product of $\vec{a}$ in equation (2),

$\vec{a}\times(\vec{a}\times\vec{c})=\vec{a}\times\vec{b}$

$(\vec{a}\cdot\vec{c})\vec{a}-(\vec{a}\cdot\vec{a})\vec{c}=\vec{a}\times\vec{b}$ ...(3)

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 1\\ 3 & -3 & 3\end{vmatrix}$$

$=\hat{i}(6+3)-\hat{j}(3-3)+\hat{k}(-3-6)$

$\vec{a}\times\vec{b}=9\hat{i}-9\hat{k}$

$\vec{a}\cdot\vec{a}=(\hat{i}+2\hat{j}+\hat{k})\cdot(\hat{i}+2\hat{j}+\hat{k})$

= 1 + 4 + 1 = 6

Putting all these values in equation (3)

$\Rightarrow\ 3(\hat{i}+2\hat{j}+\hat{k})-6\vec{c}=9\hat{i}-9\hat{k}$

$\Rightarrow\ 6\vec{c}=3\hat{i}+6\hat{j}+3\hat{k}-9\hat{i}+9\hat{k}$

$\Rightarrow\ 6\vec{c}=-6\hat{i}+6\hat{j}+12\hat{k}$

$\Rightarrow\ \vec{c}=-\hat{i}+\hat{j}+2\hat{k}$

Now, $\vec{a}\cdot((\vec{c}\times\vec{b})-\vec{b}-\vec{c})$

$$\vec{c}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 1 & 2\\ 3 & -3 & 3\end{vmatrix}$$

$=\hat{i}(9)-\hat{j}(-9)+\hat{k}(0)$

$=9\hat{i}+9\hat{j}$

$(\vec{c}\times\vec{b})-\vec{b}-\vec{c}$

$=9\hat{i}+9\hat{j}-3\hat{i}+3\hat{j}-3\hat{k}+\hat{i}-\hat{j}-2\hat{k}$

$=7\hat{i}+11\hat{j}-5\hat{k}$

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70.

General equation of circle is

$x^2 + y^2 + 2gx + 2fy + c = 0$

As it passes through (0, 0) (0, 1) (1, 0)

$\therefore \quad c = 0$

Now $2g + 1 = 0$

$\Rightarrow \quad g = \frac{-1}{2}$

$2f + 1 = 0$

$f = \frac{-1}{2}$

Hence equation of circle is

$x^2 + y^2 - x - y = 0$

Also point $(2k, 3k)$ lies on the circle.

$\therefore \quad 4k^2 + 9k^2 - 5k = 0$

$\Rightarrow \quad 13k^2 - 5k = 0$

$\Rightarrow \quad k = 0, k = \frac{5}{13}$

$\therefore \quad k = \frac{5}{13}$

71.

$a = P(X = 3) \Rightarrow$ third toss obtains 6 at first time

$\Rightarrow P(A) = \frac{1}{6}$ (obtaining a six)

$P(\bar{A}) = \frac{5}{6}$ (not obtaining a six)

$\Rightarrow a = P(X = 3) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} = \frac{25}{216} = \frac{5^2}{6^3}$

$\Rightarrow b = P(X \geq 3) =$

$\frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots$

$\Rightarrow$ It is a G.P. with first term $= \frac{25}{216} = \frac{5^2}{6^3}$ and

common ratio $\left(\frac{5}{6}\right)$

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$$\Rightarrow \text{Sum} = \frac{\frac{25}{216}}{1-\frac{5}{6}} = \frac{25}{36} = \left(\frac{5}{6}\right)^2$$

$$c = P\left(\frac{X \geq 6}{X > 3}\right) = \frac{P(X \geq 6 \cap X > 3)}{P(X > 3)} = \frac{P(X \geq 6)}{P(X > 3)}$$

$$P(X > 3) = \frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6} + \frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6} + \ldots = \frac{\frac{5^3}{6^4}}{1-\frac{1}{5}} = \frac{5^3}{6^3}$$

$$P(X \geq 6) = \frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}\times\frac{1}{6} + \left(\frac{5}{6}\right)^6 \times\frac{1}{6} + \ldots = \frac{\frac{5^5}{6^6}}{1-\frac{5}{6}} = \left(\frac{5^5}{6^5}\right)$$

$$c = \frac{\frac{5^5}{6^5}}{\frac{5^3}{6^3}} = \frac{5^2}{6^2} = \left(\frac{5}{6}\right)^2$$

$$\frac{b+c}{a} = \frac{\frac{25}{36}+\frac{25}{36}}{\frac{25}{216}} = 2\times\frac{25}{36}\times\frac{216}{25} = 12$$

72. $\cos 2x + a\sin x = 2a - 7$

$1 - 2\sin^2 x + a\sin x = 2a - 7$

Let $\sin x = t$

$2t^2 - at + 2a - 8 = 0$

$\Rightarrow (t-2)(2t + 4 - a) = 0$

$\Rightarrow a = 2\sin x + 4$

$\therefore a \in [2, 6] \Rightarrow p = 2$ and $q = 6$.

Now

$\tan\theta + \cot\theta = 2\text{cosec}2\theta$

$r = 2\text{cosec } 18° - 2 \text{ cosec } 54°$

$$= 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right]$$

$$= 8\times\frac{2}{5-1} = 4, \ = r$$

$\therefore pqr = 2 \times 6 \times 4 = 48.$

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73.

. $f(x)-f(y)\geq \ln\left(\frac{x}{y}\right)+x-y$

Put $x = x + h$ and $y = x$, where $h > 0$

$\Rightarrow \quad f(x+h)-f(x)\geq \ln\left(\frac{x+h}{x}\right)+(x+h)-x$

$\Rightarrow \quad \frac{f(x+h)-f(x)}{h}\geq \frac{\ln\left(\frac{x+h}{x}\right)+h}{h}$

$\Rightarrow \quad f'(x)\geq \lim\limits_{h\to 0}\frac{\ln\left(1+\frac{h}{x}\right)}{h}+1$

$\Rightarrow \quad f'(x)\geq \frac{1}{x}+1 \qquad \ldots(i)$

$f(y)-f(x)\leq \ln\left(\frac{y}{x}\right)+y-x$

Put $y \to x + h$, $x \to x$ where $h > 0$

$f(x+h)-f(x)\leq \ln\left(\frac{x+h}{x}\right)+h$

$\frac{f(x+h)-f(x)}{h}\leq \frac{\ln\left(1+\frac{h}{x}\right)}{h}+1$

$\Rightarrow \quad f'(x)\leq \frac{1}{x}+1 \qquad \ldots(ii)$

From (i) and (ii),

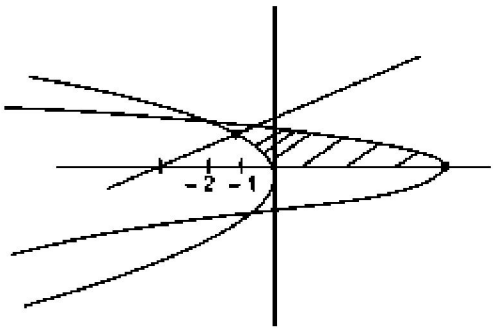
$f'(x)=1+\frac{1}{x}=\frac{x+1}{x}$

$\Rightarrow \quad f'\left(\frac{1}{n^2}\right)=n^2+1$

$\Rightarrow \quad \sum\limits_{n=1}^{20} f'\left(\frac{1}{n^2}\right)=20+\frac{20\times 21\times 41}{6}=2890$

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74.



Area between these curve will be

$$\int_{-2}^{-1}\left(-\sqrt{\frac{-x}{2}}+\frac{x+4}{2}\right)dx+\int_{-1}^{0}\left(\sqrt{\frac{8-x}{4}}-\sqrt{\frac{-x}{2}}\right)dx$$

$$+\int_{0}^{8}\sqrt{\frac{8-x}{4}}\,dx=\frac{107}{12}=\frac{m}{n}$$

$\Rightarrow m+n=119$

75.

$\because\ \alpha^2+\alpha+1=0$

$$\Rightarrow\ \alpha=\frac{-1\pm\sqrt{1-4}}{2}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$$

$\alpha=\omega,\omega^2$

If $\alpha=\omega$

$$(1+\alpha)^7=(1+\omega)^7=(-\omega^2)^7=-\omega^{14}=-\omega^2$$

$\Rightarrow\ -\omega^2=A+B\omega+C\omega^2$

$\Rightarrow\ A+B\omega+(C+1)\omega^2=0$

$\Rightarrow\ 2A-B-C=1$ and $B-C=1$

$\Rightarrow\ B=C+1,\ A=B=C+1$

$\Rightarrow\ 5(3A-2B-C)=5\ (3C+3-2C-2-C)=5$

If $\alpha=\omega^2$

$(1+\alpha)^7=(1+\omega^2)^7=-\omega^7=-\omega$

$\Rightarrow\ -\omega=A+B\omega^2+C\omega^4=A+B\omega^2+C\omega$

$\Rightarrow\ A+B\omega^2+(C+1)\omega=0$

$2A-B-C=1$ and $B-C=1$

$\Rightarrow\ B=C+1,\ A=B=C+1$

$\Rightarrow\ 5(3A-2B-C)=5$

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